

then, various new hardware platforms such as SATA hard disks, Flash disks, SATA optical drives, and various new Gigabit LAN cards have become standard. But they remain unsupported on IPCop 1.4.21. Flash-based installations are also becoming more popular than hard disk-based installations. As it is said, all good things come to an end ... or get upgraded, and so it is with IPCop. Its new version 2.0.0 was released on September 23, 2011. The latest in the series is 2.0.4, which was released on February 16, 2012.

As mentioned on IPCop.org, the latest version 2.0 has various important changes:

- Linux kernel 2.6.32
- New hardware support, including Cobalt, Sparc and PPC platforms
- A new installer, which allows you to install to Flash or hard drives, and to select interface cards and assign them to particular networks

- Access to all Web interface pages is now password protected
- The port for https secure connections has been changed to 8443
- Incorporation of OpenVPN functionality
- Several new Web interface pages

Watch this column for more about this latest IPCop release and much more. 

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(Continued from Page 80....)

Guest OS security: From a systems administrator's point of view, the virtual machines are pretty much the same as physical servers. Careful configuration and regular patching is essential for the Xen DomainU instances running the operating systems installed on them. The correct way to achieve this is to follow the OS hardening guide for the particular OS being used. This applies to the applications installed on each guest as well. It is important to remember that one vulnerable machine in a virtual environment can act as the weakest link from the security standpoint, allowing malware and viruses to spread to other guest instances, resulting in a wider security problem.

Guest communication security: Unlike physical servers where the TCP/IP communication between hosts can be sniffed and captured to detect security attacks, the communication of virtual machines is not visible on the physical network interface. This happens because the network interface card is actually being abstracted by the Xen hypervisor layer. As a result, traditional intrusion detection systems fail to detect vulnerability exploitations happening within virtual machines. The correct way to resolve this situation is to have one guest virtual machine dedicated for sniffing purposes, for all the other instances on that physical host. The data gathered thus can be fed into an IDS system to find out potential exploitations.

Host external communication security: The physical machine that hosts multiple Xen guests is going to be susceptible to network Layer-2 to Layer-7 based attacks such as eavesdropping and man-in-the-middle attacks. Implementing security solutions for the corresponding network layers such as IPSec, SSL, etc., is always a good idea. Firewalls and UTL devices to ensure the perimeter defence for a physical host are also required.

Besides these, there are a few other challenges worth taking note of. During the installation phase, it is common to have virtual guest instances created for testing purposes, which are later forgotten about and hence not removed. Such instances are usually not patched and so become a source for vulnerabilities or the weakest link. It is a best practice to delete such instances by using the Xen configuration manager. Since various guests may be running different versions and flavours of operating systems, a unified patch management solution that can deal with multiple distros is highly recommended. The same holds true for the anti-virus software. Virtual infrastructure calls for tighter physical security as well, because in very rare situations, if a physical host is stolen, it is like losing multiple servers.

Summary

While securing a virtualisation platform such as Xen, emphasis should be first given to the concepts of guest instances and how those work with the physical host. Most of the security challenges in Xen can be mitigated by proper configuration management, the deployment of best practices and security patching. It is important for today's network administrators to know about the various open source virtualisation platforms and the methods to secure their environments. 

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